A Band Divide-and-Conquer Multispectral and Hyperspectral Image Fusion Method

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***Abstract*— The nonoverlapped spectrum range between low spatial resolution (LR) hyperspectral (HS) and high spatial resolution (HR) multispectral (MS) images has been a fundamental but challenging problem for MS/HS fusion. The spectrum of HS data is generally 400–2500 nm, and the spectrum of MS data is generally 400–900 nm; how to obtain the high-fidelity HR HS fused image within the whole spectrum of 400–2500 nm? In this article, we proposed a band divide-and-conquer framework (BDCF) to solve the problem, by comprehensively considering spectral fidelity, spatial enhancement, and computational efficiency. First, the spectral bands of HS were divided into overlapped and nonoverlapped bands according to the spectral response between HS and MS. Then, a novel improved component substitution (CS)-based method by combing neural network was proposed to fuse the overlapped bands of LR HS. Then, a mapping-based method with the neural network was presented to construct the complicated nonlinear relationship between overlapped and nonoverlapped bands of the original LR HS data. The trained network was mapped to the fused overlapped HR HS bands to estimate the nonoverlapped HR HS bands. Experimental results on two simulated data sets and two realistic data sets of Gaofen (GF)-5 LR HS, GF-1 MS, and Sentinel-2A MS show that the proposed BDCF has superior performance in both high spectral fidelity and sharp spatial details, and it obtained competitive fusion behaviors compared with other state-of-the-art methods. Moreover, BDCF has relatively higher computational efficiency than optimal solution-based methods and deep learning-based fusion methods.**

***Index Terms*— Divide-and-conquer framework (BDCF), Gaofen-5 (GF-5), hyperspectral (HS), image fusion, multispectral (MS), neural network framework.**

I. Introduction

DUE to the tradeoff of spatial resolution, spectral resolution and signal-to-noise ratio (SNR) on the designed sensor, satellite remote sensing images with both high spatial resolution and high spectral resolution could not be perfectly obtained [1], [2]. The hyperspectral (HS) images generally have over 100 bands with abundant spectral information [3]; however, its spatial resolution is relatively low. On the contrary, multispectral (MS) image has limited bands but relatively high spatial resolution [4]. Therefore, many scholars have been devoted to fusing high resolution (HR) MS and low resolution (LR) HS to obtain the HR HS data.

Current fusion methods of HS and MS can be roughly grouped into five categories [5]: component substitution (CS)-based methods [6], multiresolution analyze (MRA)-based methods [7], [8], unmixing-based methods [9], Bayesian-based methods [10], [11], and deep learning-based methods [12], [13]. CS-based and MRA-based methods originate from pan-sharpening approaches [14], [15] CS-based methods achieve image fusion by replacing the intensity component of HS with MS, and different algorithms implement divergent forward transform coefficients and gain matrices [6]. Typical algorithms are intensity–hue–saturation (IHS) [16], adaptive Gram–Schmidt (GSA) [17] and principal component analysis (PCA) [18]. In MRA-based methods, the spatial details of MS are extracted and injected into HS to realize the image fusion. Their fusion behaviors rely on selected filters and gain matrices [19]. Typical algorithms include decimated wavelet transform (DWT) [20], "a Ì trous" wavelet transform (ATWT) [21] and modulation transfer functions-generalized Laplacian pyramid (MTF-GLP) [5], [22]. These traditional pan-sharpening methods have high computational efficiency; however, it lacks in fusion quality [6]. For example, CS-based methods often suffer from spectral distortions due to CS that alters the spectral characteristics of the original data. MRA-based methods achieve good spectral fidelity, but these methods are limited for the enhancement of spatial resolution. Unmixing-based methods decompose HS and MS data into endmember and abundance matrices. The fused image is obtained by reconstructing the HS endmember matrix and the MS abundance matrix [23]. Typical algorithms are

Manuscript received September 11, 2020; revised November 10, 2020 and December 8, 2020; accepted December 12, 2020. Date of publication February 25, 2021; date of current version December 6, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 41971296, Grant 41671342, Grant 41801252, Grant 41801256, Grant U1609203, and Grant 61871177; in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LR19D010001 and Grant LQ18D010001; and in part by the Natural Science Foundation of Ningbo City under Grant 2019A610098. *(Corresponding authors: Xiangchao Meng; Jiangtao Peng.)*

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Digital Object Identifier 10.1109/TGRS.2020.3046321

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coupled nonnegative matrix factorization (CNMF) [24] and Lanara's method [25]. Bayesian-based methods maximize the posterior probability density with prior constrains to achieve image fusion [10]. These methods include fast fusion based on Sylvester equation (FUSE) [26] and maximum a posterior-stochastic mixing model (MAP-SMM) [5]. Recently, deep learning-based methods have been popular in many study areas and obtained good performance in image fusion. These methods formulate the mapping relationships between LR HS, HR MS and HR HS by establishing different networks, and then map this relationship to another data set (LR HS and HR MS that need to be fused) to achieve image fusion [27], [28]. Typical examples are 3-D-convolutional neural network [29], deep two-branch convolutional neural network [30], and pyramid fully convolutional network [31]. Unmixing-based methods, Bayesian-based methods, and deep learning-based methods achieved comparative fusion results; however, there involve many parameters, and the computational efficiency is relatively low [5], [32]. There still exist two severe issues to address.

1) The performance of current fusion methods is limited by nonoverlapped spectrum range between HS and MS images. As illustrated in Fig. 1, the spectral ranges of HS and MS data are generally 400–2500 and 400–900 nm, respectively, and their divergences are obvious. However, almost all state-of-the-art methods did not carefully explore the problem. The typical CNMF method achieved good fusion performance in the overlapped bands in Fig. 1, but it exhibited poor behaviors within the nonoverlapped bands, which would severely degrade its specific application performance. For example, the spectral distortions of bands between 1550–1750 and 2080–2350 nm [33] bring about distinct calculation errors of normalized difference building index, and would negatively affect the mapping results of urban investigation [5], [34].
2) Most of current methods could not simultaneously obtain good fusion performance and high computational efficiency. As shown in Fig. 2, the Bayesian-based methods, unmixing-based methods, and deep learning-based methods have the superiority of spatial sharpening and spectral fidelity 5], [6], but they are generally time-consuming. At the same time, the CS-based methods and MRA-based method have higher computational efficiency [35], but they are not robust enough and the spectral or image quality of the fused image is generally degraded. Accordingly, most of methods are not perfect for handling realistic fusion problems.

Aiming at the above two problems, in this article, we presented a band divide-and-conquer framework (BDCF) to fuse LR HS and HR MS data. This idea of our method is to divide spectral bands of HS into overlapped and nonoverlapped bands according to the spectral range of HS and MS data. For the overlapped bands, a novel improved CS-based method by combing neural network was presented. For the nonoverlapped bands, a mapping based method with the neural network was proposed to explore the nonlinear relations between

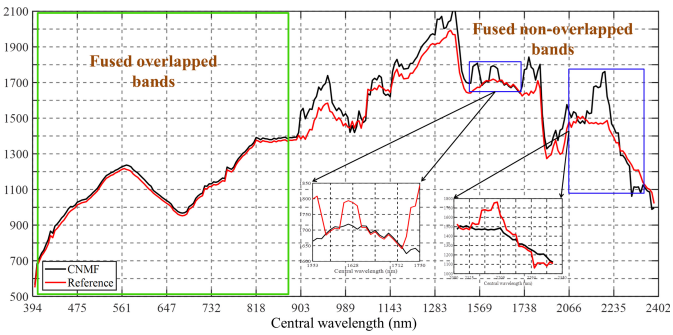


Fig. 1. Difference of fusion quality for overlapped and nonoverlapped bands based on the CNMF method.

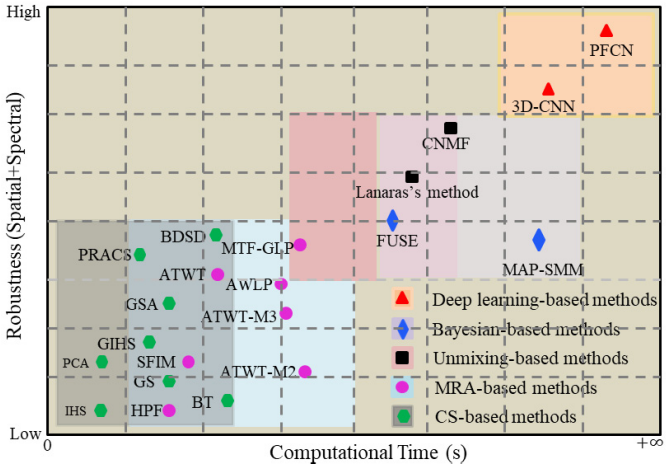


Fig. 2. Robustness and computational time of current fusion methods.

overlapped and nonoverlapped bands of the original HS. Then, the nonlinear relationship was mapped into the fused overlapped bands, to predict the HR nonoverlapped bands. We finally combined the fused results of overlapped and nonoverlapped bands to achieve the aimed HR HS. The main contributions of this article are as follows.

1) Our BDCF stands on the idea of divide-and-conquer to handle the MS/HS fusion issue by separately fusing the overlapped and nonoverlapped bands between LR HS and HR MS data. It could effectively solve the spectral distortion problem of the nonoverlapped bands between the HS and MS data.
2) Our BDCF proposed two innovative schemes to guarantee the MS/HS fusion performance. It presented an improved CS-based algorithm by combing neural network for overlapped HS band fusion. The improved CS-based algorithm could predict the optimal HR intensity component through neural network, and improves the spectral fidelity of the fused image while enhancing its spatial resolution. Moreover, our BDCF adopted a novel strategy for fusing nonoverlapped HS bands, by exploring the nonlinear relationship between overlapped and nonoverlapped bands.
3) Our BDCF is easy to implement and effective, and achieve competitive results with high spatial sharpness, high spectral fidelity, and acceptable computing efficiency.

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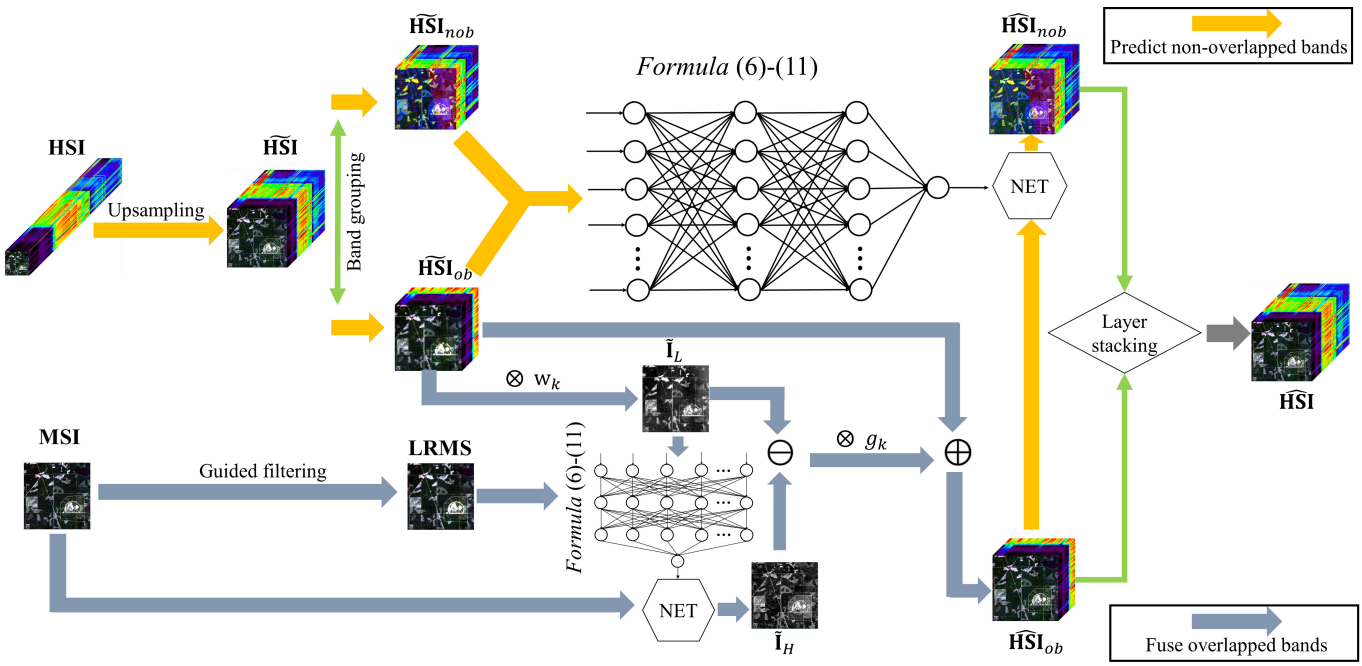


Fig. 3. Flowchart of BDCF method.

The structure of this article is organized as follows. The proposed fusion method is specifically described in Section II. The experimental results and analysis are presented in Section III. The conclusion is given in Section IV.

II. BDCF

A. Idea of BDCF

HS and MS data always have some differences in spectral range, and the characteristics and concerned fusion problems for overlapped and nonoverlapped bands of HS are different [36]. Based on the above motivations, we proposed the BDCF for HS and MS fusion, as is shown in Fig. 3. We use bold uppercase to represent matrices (e.g., **W**) and lowercase letters to express vectors (e.g., w). Thus, $\mathbf{HSI} \in \mathbb{R}^{n\times m\times B}$ denotes the LR HS consisting of $n \times m$ pixels and B bands. $\mathbf{MSI} \in \mathbb{R}^{N\times M\times b}$ is the observed HR MS data with $N \times M$ pixels and b bands, where $n < N, m < M$, and $B > b$. The LR HS data upsampled to the MS scale is denoted by $\widetilde{\mathbf{HSI}} \in \mathbb{R}^{N\times M\times B}$, and the fused image is denoted by $\widehat{\mathbf{HSI}} \in \mathbb{R}^{N\times M\times B}$. Besides, the subscript ob denotes the overlapped bands (these bands of MS and HS data in 400–900 nm) and nob is the nonoverlapped bands (these MS bands of 400–900 nm and HS bands of 900–2500 nm).

The BDCF consists of two parts: overlapped band fusion and nonoverlapped band prediction. The overlapped bands and nonoverlapped bands can be obtained by spectral matching between HS and MS data. For the overlapped bands, an improved CS-based method by combing neural network was proposed. First, we extracted the intensity component of HS data. Then, we proposed to implement the neural network model for the optimal matching between intensity component of LR HS and filtered MS data, to obtain the optimal HR intensity component. Finally, we replaced the intensity component of LR HS data with the HR intensity component to realize the high-fidelity fusion of overlapping bands images. For the nonoverlapped bands, we first implemented the neural network to explore nonlinear relations between overlapped and nonoverlapped bands of original HS data. Then, we predicted the HR nonoverlapping bands by mapping the nonlinear relationship into the fused overlapped bands. After the abovementioned procedure, we combined the fused results of overlapped and nonoverlapped bands to achieve the final HR HS.

B. Fusing Overlapped Bands

CS-based methods have high computational efficiency and obvious spatial resolution enhancement effect. However, the traditional CS-based method uses HR MS to replace the LR intensity component of HS data to achieve fusion. The extremely low correlation between HR MS and LR intensity components results in the change of spectral characteristics of HS data and serious spectral distortion. Therefore, an appropriate intensity component is very important for component replacement of HS data. In our method, we aimed to solve this problem, and an improved CS-based method by combing neural network was proposed. The relationship between LR MS and LR intensity components is trained by neural network algorithm, and the HR intensity component is obtained by mapping the network to HR MS in the form of end-to-end. Our method can improve the spatial resolution of HS image while better preserving the spectral characteristics. The detailed procedure is as follows.

First, the intensity component is extracted from the overlapped bands of the upsampled LR HS, shown as follows:

$$\tilde{\mathbf{I}}_L = \sum_{k=1}^{\theta} w_k \widetilde{\mathbf{HSI}}_{\mathrm{ob}}^{k} \tag{1}$$

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where θ is the number of the overlapped LR HS bands, and $\mathbb{K}$ is the $\mathbb{K}$th spectral band. $\tilde{\mathbf{I}}_L \in \mathbb{R}^{N\times M}$ is the LR intensity component. w_k indicates the weight of $\mathbb{K}$th band and is calculated by using the following equations:

$$\mathbf{P} = \frac{\mathbf{MSI}}{b} \tag{2}$$

$$\mathbf{P}_L = \sum_{k=1}^{\theta} w_k \widetilde{\mathbf{HSI}}_{\text{ob}}^{k} \tag{3}$$

where $\mathbf{P}$ is the average image of MS. $\mathbf{P}_L \in \mathbb{R}^{N\times M}$ is the filtered $\mathbf{P}$ by using the Gaussian filter, and the w_k is obtained by calculating the minimal mean squared error of (3).

Then, each band of MS image is filtered according to the extracted intensity component $\tilde{\mathbf{I}}_L$ by guided filtering [37]

$$\overline{\overline{\mathbf{MSI}}}_t^k = a_z \tilde{\mathbf{I}}_{L_t} + b_z, \quad t \in W_z \tag{4}$$

where W_z is fixed-size sliding zth window, $\tilde{\mathbf{I}}_{L_t}$ represents the tth pixel of $\tilde{\mathbf{I}}_L$, $\overline{\overline{\mathbf{MSI}}}_t \in R^{N\times M\times b}$ is the tth pixel of the filtered MS. The coefficients a_z and b_z are obtained by solving the following linear regression:

$$E(a_z, b_z) = \sum_{t\epsilon w_z} \left(\left(a_z \tilde{\mathbf{I}}_{L_t} + b_z - \mathbf{MSI}_t^k \right)^2 + \varepsilon a_z^2 \right) \tag{5}$$

where $\mathbf{MSI}_t$ represents the tth pixel of $\mathbf{MSI}$, and ε is a regularization parameter.

Due to deep learning-based methods have extremely low computational efficiency, a simple neural network framework was built in this article, and it has higher computational efficiency and is more practical. The neural network includes input layer, hidden layer and output layer [38]. The number of neurons of input layer and hidden layers is set manually, and the number of neurons of output layer is 1 for simplicity. The output a_j of the hidden layer neuron j is defined as

$$\mathbf{a}_j = f\left(\sum_i w_{ji} \cdot \mathbb{K}_i + b_j \right) \tag{6}$$

where $\mathbb{K}$ is the input data, $\mathbb{K}_i$ is the ith neuron of the $\mathbb{K}$, w_{ji} is the weight from ith neuron of the input layer to jth neuron of the hidden layer, b_j is the bias, and $f(\cdot)$ uses the sigmoid function.

Then, the output value $\mathbf{A}_p$ of the jth neuron of the output layer can be obtained from the following equation:

$$\mathbf{A}_p = f\left(\sum_j w_j \cdot a_j + b \right) \tag{7}$$

where w_j is the weight from the jth neuron of the hidden layer to the output layer, and b is the bias.

The weights and the bias are successively updated in the neural network framework using the following equations:

$$w_j = w_j + \alpha \cdot \left(\mathbf{A}_p - \mathbf{A}_r \right) \cdot \mathbf{A}_p \cdot \left(1 - \mathbf{A}_p \right) \cdot a_j \tag{8}$$

$$b = b + \beta \cdot \left(\mathbf{A}_p - \mathbf{A}_r \right) \cdot \mathbf{A}_p \cdot \left(1 - \mathbf{A}_p \right) \tag{9}$$

$$w_{ji} = w_{ji} + \alpha \cdot \left(\mathbf{A}_p - \mathbf{A}_r \right) \cdot w_j \cdot a_j \cdot \left(1 - a_j \right) \cdot \mathbb{K}_i \tag{10}$$

$$b_j = b_j + \beta \cdot \left(\mathbf{A}_p - \mathbf{A}_r \right) \cdot w_j \cdot a_j \cdot \left(1 - a_j \right) \tag{11}$$

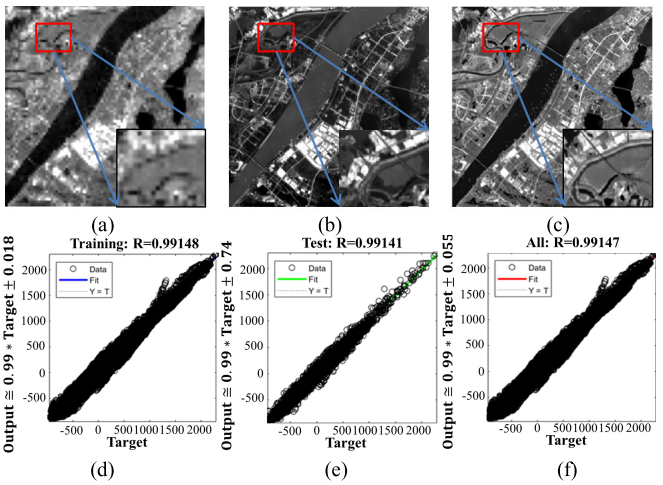


Fig. 4. Comparison of intensity component obtained by traditional method and the proposed method. (a) Intensity component of LR HS. (b) HR intensity component in traditional CS-based methods. (c) Predicted HR intensity component of the proposed method. (d) Fit accuracy of the proposed HR intensity component and the intensity component of LR HS to be submitted based on the training data set. (e) Fit accuracy of the proposed HR intensity component and the intensity component of LR HS to be submitted based on the test data set. (f) Overall fit accuracy of the proposed HR intensity component and the intensity component of LR HS to be submitted.

where $\alpha = 0.6$ and $\beta = 0.7$ are the parameters, and $\mathbf{A}_r$ is the target data.

More training samples in the neural network framework usually achieve better training results. Therefore, using $\overline{\overline{\mathbf{MSI}}}$, not $\mathbf{P}_L$, as input data and $\tilde{\mathbf{I}}_L$ as output data, (6)–(11) is executed to obtain the mapping relationship between the filtered MS and the LR intensity component. The HR intensity component $\tilde{\mathbf{I}}_H$ is obtained by mapping the relationship to MS

$$\tilde{\mathbf{I}}_H = \left(\mathbf{MSI}_i \cdot w_{ji} + b_j \right) \cdot w_j + b \tag{12}$$

where $\mathbf{MSI}_i$ is the ith neuron of MS data. Fig. 4 shows the mapped intensity component and the performance of the neural network framework, and the predicted result is more suitable for CS.

Finally, the fused image is obtained by using the following equation:

$$\widehat{\mathbf{HSI}}_{\text{ob}}^k = \widetilde{\mathbf{HSI}}_{\text{ob}}^k + \frac{\text{cov}\left(\tilde{\mathbf{I}}_L, \widetilde{\mathbf{HSI}}_{\text{ob}}^k \right)}{\text{var}\left(\tilde{\mathbf{I}}_L \right)} \left(\tilde{\mathbf{I}}_H - \tilde{\mathbf{I}}_L \right) \quad k = 1, \ldots, \theta \tag{13}$$

where $\text{cov}(\cdot)$ represents the covariance between two images, and $\text{var}(\tilde{\mathbf{I}}_L)$ is the variance of $\tilde{\mathbf{I}}_L$.

C. Predicting Nonoverlapped Bands

The MS and nonoverlapped bands of HS data have extremely low correlation, which results in poor fusion quality of nonoverlapped bands, and how to solve this problem is not considered in current fusion methods. In this article, aims to solve this problem, a mapping-based method with the neural network was proposed to explore the nonlinear relations between overlapped and nonoverlapped bands of the original HS. The nonlinear relationship between overlapped and nonoverlapped bands of original HS is trained by the neural

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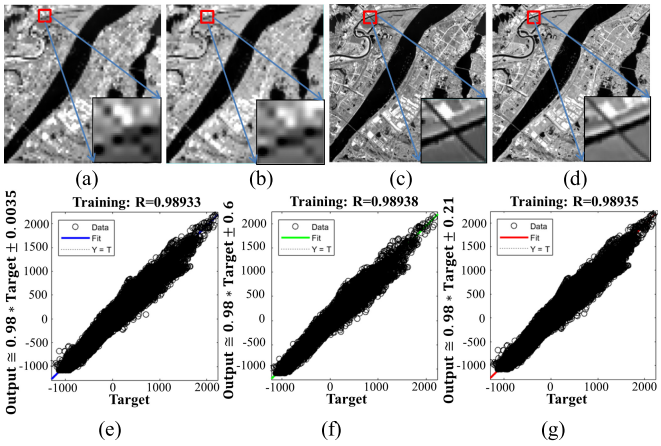


Fig. 5. Performance of the proposed mapping-based method for nonoverlapped band fusion. (a) Principal component of the original overlapped bands $\widehat{\mathbf{HSI}}_{\mathrm{ob}}$ (input of the network training). (b) Principal component of the original nonoverlapped bands $\widehat{\mathbf{HSI}}_{\mathrm{nob}}$ (output of the network training). (c) Principal component of fused overlapped bands $\widehat{\mathbf{HSI}}_{\mathrm{ob}}$ (input of the network testing). (d) Principal component of predicted nonoverlapped bands $\widehat{\mathbf{HSI}}_{\mathrm{nob}}$ (output by the proposed training network). (e) Fit of training data set. (f) Fit of test data set. (g) Overall fit.

network model, and the well-trained nonlinear relationship is then mapped to the fused overlapped bands to obtain the HR nonoverlapped bands. To improve the computational efficiency and obtain better fusion quality, the principal components of the overlapped, nonoverlapped of the original HS and the overlapped bands of the fused image were first calculated, respectively, using the following equations:

$$\left[\widetilde{\mathbf{coeff}}_{\mathrm{ob}}, \widetilde{\mathbf{PC}}_{\mathrm{ob}}\right] = \mathrm{PCA}\left(\widetilde{\mathbf{HSI}}_{\mathrm{ob}}\right) \tag{14}$$

$$\left[\widetilde{\mathbf{coeff}}_{\mathrm{nob}}, \widetilde{\mathbf{PC}}_{\mathrm{nob}}\right] = \mathrm{PCA}\left(\widetilde{\mathbf{HSI}}_{\mathrm{nob}}\right) \tag{15}$$

$$\left[\widehat{\mathbf{coeff}}_{\mathrm{ob}}, \widehat{\mathbf{PC}}_{\mathrm{ob}}\right] = \mathrm{PCA}\left(\widehat{\mathbf{HSI}}_{\mathrm{ob}}\right) \tag{16}$$

where $\widetilde{\mathbf{coeff}}$ is the forward transform coefficient of PCA of data, $\mathbf{PC}$ is the principal component of data.

Then, the nonlinear relationship of the principal component between overlapped (input data) and nonoverlapped bands (output data) of original HS is trained using (6)–(11), and the predicted principal component of HR nonoverlapped bands is obtained by the following equation:

$$\widehat{\mathbf{PC}}_{\mathrm{nob}} = \left(\widehat{\mathbf{PC}}_{\mathrm{ob}_i} \cdot w_{ji} + b_j\right) \cdot w_j + b \tag{17}$$

where $\mathbf{PC}$ represents the principal component, $\widehat{\mathbf{PC}}_{\mathrm{ob}_i}$ is the ith neuron of $\widehat{\mathbf{PC}}_{\mathrm{ob}}$. Fig. 5 shows the mapped first principal component of $\widehat{\mathbf{HSI}}_{\mathrm{nob}}$ and the performance of the neural network framework, and the mapped first principal component of $\widehat{\mathbf{HSI}}_{\mathrm{nob}}$ obtained better spectral fidelity and sharp spatial details.

Finally, the spatial resolution of nonoverlapped bands is enhanced by replacing the principal component and the other components remain unchanged, and performing the inverse PCA transform using the following equation:

$$\widehat{\mathbf{HSI}}_{\mathrm{nob}} = \widehat{\mathbf{PC}}_{\mathrm{nob}} \cdot \mathbf{coeff}^{T}_{\mathrm{nob}} \tag{18}$$

where T stands for inverse operation.

III. Experiments

A. Experimental Data Sets

Four data sets were implemented to verify the performance of BDCF. The data sets include two simulated data sets and two realistic data sets, with detailed information shown in Table I.

All simulated data sets are from Gaofen-5 (GF-5) HS sensor, the spectral range is 400–2500 nm, and the spatial resolution is 30 m. "Langfang" [see Fig. 6(c)] data set covers part of Langfang City, Hebei Province, China and was captured on October 14, 2018. The spatial size of HS data is 400 × 400. We discarded the 39 bands of water absorption, and the final number of spectral bands used was 291. "Poyang Lake-1" [see Fig. 7(c)] covers part of Poyang Lake, China, on October 7, 2018, and consists of 450 × 450 pixels. The total of 62 bands of water absorption were removed, and 268 bands were used in the experiment.

The abovementioned HS images were used as the reference image. The obtained LR HS [see Figs. 6(a) and 8(a)] and the HR MS [see Figs. 6(b) and 8(b)] were generated by the following three steps. First, to simulate the observed MS, the spectrum of referenced images was downsampled by the spectral response functions. In these experiments, the spectral response functions of GF-1 and GeoEye-1 sensors were selected, which correspond to "Langfang" and "Poyang Lake-1," respectively. Second, to simulate the observed HS, the reference images were spatially blurred using the Gaussian filter as the blur kernel and down-sampled by factors. The factors of "Langfang" and "Poyang Lake-1" were 4 and 5, respectively. Finally, the Gaussian noise was added to the simulated HS and MS.

All realistic data sets of HS data are also from GF-5 HS sensor. "Poyang Lake-2" [see Fig. 11 HS and MS] covers part of Poyang Lake area. This HS data is 400 × 400 pixels and captured on October 7, 2018. We removed 46 bands of water absorption, and 284 bands were used in the experiment. Sentinel-2A MS acquired on October 4, 2018, is 1200 × 1200 × 4 pixels, its spatial resolution and spectral range are 10 m and 400–900 nm, respectively. "Taihu Lake" [see Fig. 12 HS and MS] covers part of Taihu Lake area, China. This HS data consists of 250 × 250 pixels and was acquired on June 1, 2018. Only 45 bands were removed, and the number of bands used was 285. GF-1 sensor captured the MS data of 1000 × 1000 pixels on June 25, 2018. It has four bands with spatial resolution of 8 m, and the spectral range is 400–900 nm. The number of neurons was set to 30. The number of PCs was set to 1 in Langfang and realistic data sets, and the number of PCs was 2 in Poyang Lake-1.

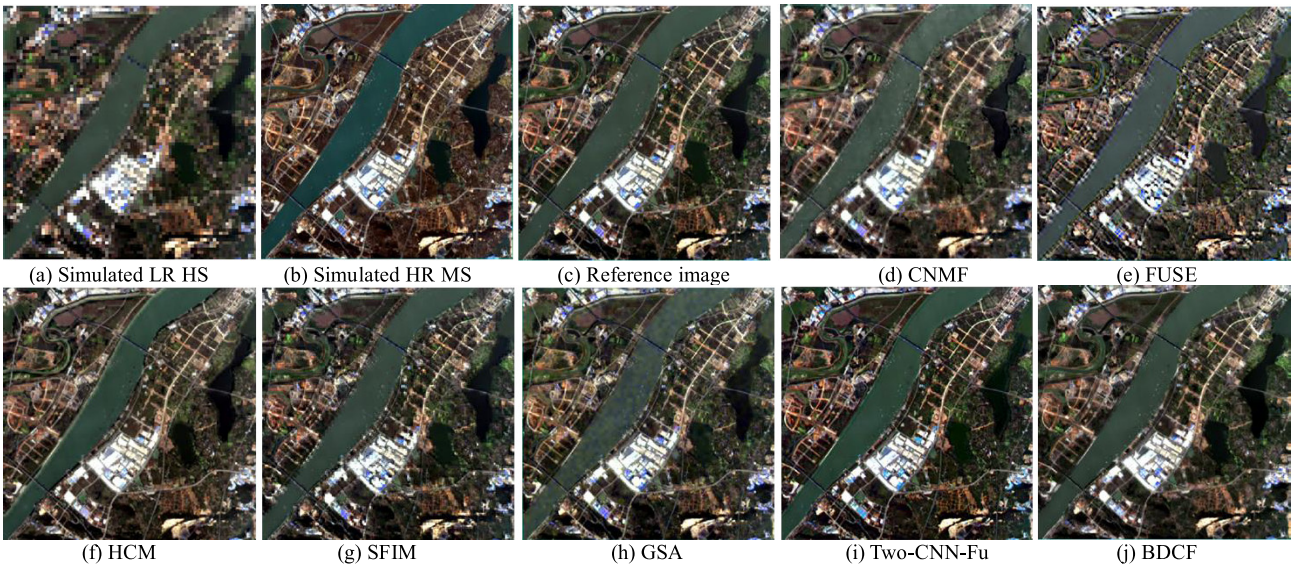
B. Evaluation Metrics

According to the Wald's protocol, the fused image should be consistent with the realistic HS data. Therefore, the fused images were compared with the reference images. Let $\mathbf{z} \in \mathbb{R}^{\mathbf{B}\times\mathbf{D}}$ represents the reference HS with B bands and $D = N \times M$ pixels, $\mathbf{z}_j \in \mathbb{R}^{B\times 1}$ and $\hat{\mathbf{z}}_j \in \mathbb{R}^{B\times 1} (j = 1, \ldots, D)$ represent spectral signatures of the jth pixel in the reference images and the fused images, respectively, and $\mathbf{z}_i \in \mathbb{R}^{D\times 1}$ is

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TABLE I
CHARACTERISTIC OF SIMULATED DATA SETS AND REAL DATA SETS

Categories	Dataset	Instrument	Semi Synthetic	Dimensions	Bands
Simulated Datasets	Langfang	GF-5	HS	80*80	291
			MS (GF-1)	320*320	4
	Poyang Lake-1	GF-5	HS	150*150	268
			MS (GeoEye-1)	750*750	4
Realistic Datasets	Poyang Lake-2	GF-5	HS	400*400	284
		Sentinel-2A	MS	1200*1200	4
	Taihu Lake	GF-5	HS	250*250	285
		GF-1	MS	1000*1000	4



(a) Simulated LR HS (b) Simulated HR MS (c) Reference image (d) CNMF (e) FUSE
(f) HCM (g) SFIM (h) GSA (i) Two-CNN-Fu (j) BDCF

Fig. 6. Fusion results in Langfang data set from different methods. (a) Simulated LR HS. (b) Simulated HR MS. (c) Reference image. (d) CNMF. (e) FUSE. (f) HCM. (g) SFIM. (h) GSA. (i) Two-CNN-Fu. (j) BDCF.

the ith band ($i = 1, \ldots, B$). To quantitative evaluate the fusion results, the following four evaluation metrics are used.

1) The spectral angle measure (SAM)

$$\mathrm{SAM}(\mathbf{z}_j, \hat{\mathbf{z}}_j) = \arccos\left(\frac{\mathbf{z}_j^t \hat{\mathbf{z}}_j}{\|\mathbf{z}_j\|_2 \|\hat{\mathbf{z}}_j\|_2}\right). \tag{19}$$

A larger SAM means more severe spectral distortions of fused images [39]. When the SAM equals to 0, the fused images have smallest spectral distortions.

2) Erreur relative globale adimensionnelle de synth $\grave{e}$se (ERGAS)

$$\mathrm{ERGAS}(\mathbf{z},\hat{\mathbf{z}}) = 100p\sqrt{\frac{1}{B}\sum_{i=1}^{B}\frac{\|\hat{\mathbf{I}}\mathbf{z}_i - \hat{\mathbf{z}}_i\|\,\hat{\mathbf{I}}_2^2}{\left(\frac{1}{D}1_D^t \mathbf{z}_i\right)^2}} \tag{20}$$

where p is the ratio of spatial resolution of the data set. A larger ERGAS brings about more spectral distortions [40]. When ERGAS is equal to 0, the spectral distortion is the smallest.

3) Peak signal-to-noise ratio (PSNR)

$$\mathrm{PSNR}(\mathbf{z}_i, \hat{\mathbf{z}}_i) = 10 \cdot \log_{10}\left(\frac{\max(\mathbf{z}_i)^2}{\|\hat{\mathbf{I}}\mathbf{z}_i - \hat{\mathbf{z}}_i\|\,\hat{\mathbf{I}}_2^2/D}\right) \tag{21}$$

where $\max(\mathbf{z}_i)$ is the maximum value in the ith band of the reference image [41]. A higher PSNR means a better result.

4) Cross correlation (CC)

$$\mathrm{CC}(\hat{\mathbf{z}}, \mathbf{z}) = \frac{1}{B}\sum_{i=1}^{B}\mathrm{CCS}(\hat{\mathbf{z}}_i, \mathbf{z}_i) \tag{22}$$

where CCS is the CC between two single-band images $\mathbf{A}$ and $\mathbf{B}$, estimated from the following equation:

$$\mathrm{CCS}(\boldsymbol{A}, \boldsymbol{B}) = \frac{\sum_{j=1}^{n}\left(\boldsymbol{A}_j - \mu_A\right)\left(\boldsymbol{B}_j - \mu_B\right)}{\sqrt{\sum_{j=1}^{n}\left(\boldsymbol{A}_j - \mu_A\right)^2 \sum_{j=1}^{n}\left(\boldsymbol{B}_j - \mu_B\right)^2}} \tag{23}$$

where j denotes the jth pixel of image, and μ_A and μ_B represent the sample mean of $\mathbf{A}$ and $\mathbf{B}$, respectively. n is the number of pixels. The ideal value is 1 [42].

C. Experimental Results

The proposed BDCF was compared with seven state-of-the-art methods in different SNRs, including CNMF [24], FUSE [26], hybrid color mapping (HCM) [43], smoothing filter-based intensity modulation (SFIM) [44], GSA [17], two-branches convolutional neural network-fully (Two-CNN-Fu) [30], pramid fully convolution network (PFCN) [31]. In addition, to more comprehensively verify the proposed method, the proposed improved CS-based method by combing neural network (CS-NN) for overlapped bands is implemented to fuse

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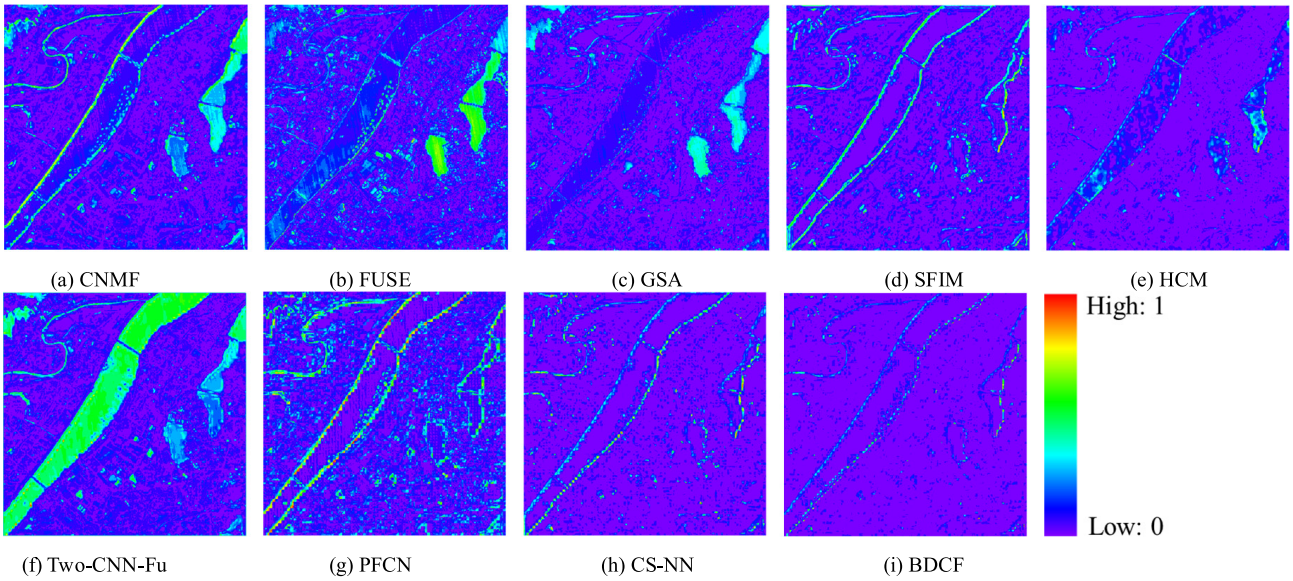


Fig. 7. SAM map in Langfang data set. (a) CNMF. (b) FUSE. (c) GSA. (d) SFIM. (e) HCM. (f) Two-CNN-Fu. (g) PFCN. (h) CS-NN. (i) BDCF.

all the spectral bands. False color images of bands at 638, 548, and 471 nm were selected for visual comparison.

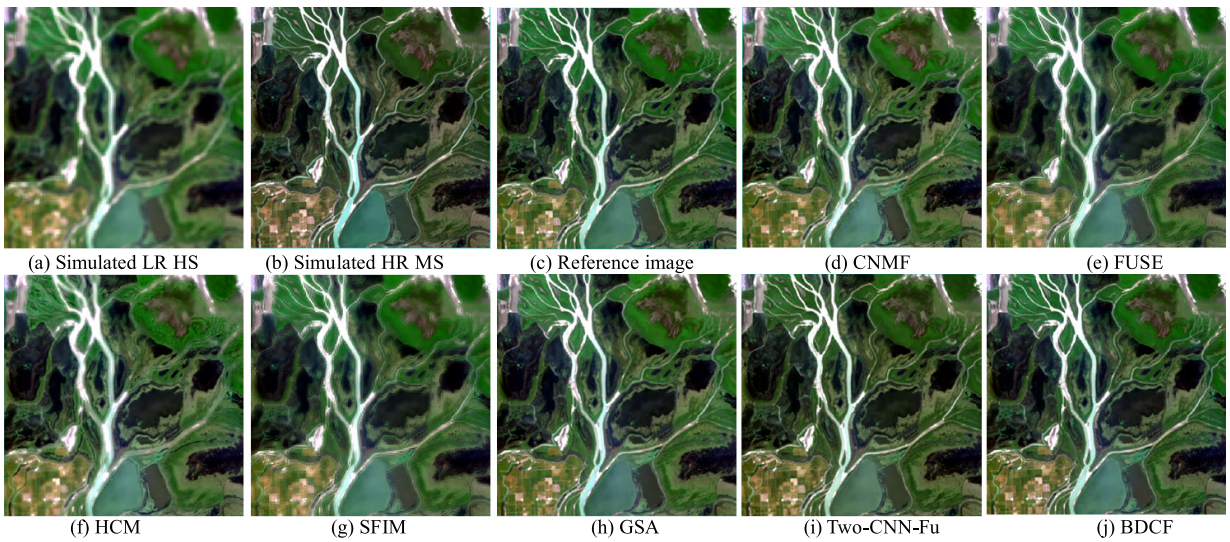
1) Simulated Data Sets Results: The fusion results of simulated data sets (SNR = 40 dB) are shown in Figs. 6 and 8, and Figs. 7 and 9 show the SAM map and the spectral error of the fused image. The spectral distortion gradually increases from 0 to 1, and the color changes from blue to red in these SAM maps. It can be seen that HCM achieves poor fusion performance, some edges with river are over enhanced, and produces some spectral distortions, particularly in the river and vegetation area. FUSE provides poor visualization results and there are obvious spectral and spatial differences with the reference image. Such as some spectral differences can be found in some areas with river and vegetation, and the edges of ground objects are blurred. Compared with the reference image, the fused images of GSA and Two-CNN-Fu have obvious spatial details and some spectral differences can be found in some area with river and vegetation. The fused results of SFIM have the closest spectral characteristics with the reference image; however, the injection of spatial details is insufficient in all data sets. CNMF obtains comparative spatial sharpening effects in all data sets and achieves poor spectral fidelity in the river and vegetation area. PFCN and CS-NN achieve better visualization results than other benchmark methods. In the proposed fused images, the spectral characteristics are better preserved and clear spatial structures can be observed. The fusion results of different SNR data sets also show that FUSE, GSA and Two-CNN-Fu are sensitive to noise data, and BDCF and other methods are robust. Table II shows the band-averaged values of quantitative evaluation of overlapped bands (ob), nonoverlapped (nob) bands and all bands (the bold type represents the best result, and the second-best result is to be underlined). BDCF gives the best values in all the indicators and the value of these nonoverlapped bands has obvious comparative advantage. Besides, Fig. 10 shows the PSNR and CC value of each band in the fused images. For PSNR, the results of CNMF, GSA and PFCN achieve higher value in each band, followed by FUSE, CS-NN, HCM and SFIM, Two-CNN-Fu is the worst, and BDCF is best in almost each band. For CC, the results of Two-CNN-Fu, GSA, PFCN, CNMF and CS-NN are better in each band, other benchmark methods are poor, particularly in FUSE and SFIM, and our method obtains the best results in each band.

2) Real Data Sets Results: Figs. 11 and 12 show the fusion results of the real data sets. Although all the fusion methods obtain sharp and clear spatial detail compared with the LR HS, yet there are still some spectral distortions in these fused images. For example, CNMF, HCM, and GSA produce some spectral distortions, particularly in farmland and vegetation area, the spectral characteristics of FUSE are changed in water area, and Two-CNN-Fu produces serious spectral distortions in the whole images. SFIM, PFCN, and CS-NN achieve comparative fusion results, and the best performance of spatial structures and spectral fidelity can be observed in the fusion result of BDCF. To fairly compare the performance of our method in spectral preservation, Figs. 13 and 14 plot the spectral curves of removed the continuum from different objects (river, vegetation, bare land, and pond). Compared with the reference curve, there are some differences of curve in these methods. From the Poyang Lake-2 data sets, in river area [see Fig. 13(a)], CNMF has a wrong valley value around 600 nm. SFIM lacks a peak value between approximately 760–900 nm. Except for BDCF, the other methods have violent fluctuations in 2007–2513 nm. Besides, the spectral curve of BDCF is almost consistent with the reference spectral curve. For the vegetation [see Fig. 13(b)], these benchmark methods have some fluctuations of spectral curves in 390–514 nm, for example, HCM lacks a valley value, and other methods have a trend similar to the reference spectral curve. However, the differences are obvious, and our method still maintains the best results. For bare land [see Fig. 13(c)], there are serious

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TABLE II
QUANTITATIVE EVALUATION OF THE FUSION RESULTS FOR THE SIMULATED EXPERIMENTS (SNR = 40 DB)

Methods	SAM		ERGAS		PSNR		CC	
	Langfang	Poyang Lake-1	Langfang	Poyang Lake-1	Langfang	Poyang Lake-1	Langfang	Poyang Lake-1
CNMF	6.9	7.0	26.7	27.6	28.5	27.8	0.96	0.95
FUSE	7.2	6.7	29.9	30.2	25.6	29.4	0.89	0.95
HCM	6.6	8.8	26.5	27.1	28.8	24.4	0.91	0.87
SFIM	6.1	6.0	25.8	25.5	31.1	31.9	0.91	0.89
GSA	7.5	6.4	30.	31.6	24.8	30.9	0.95	0.95
Two-CNN-Fu	12.9	13.7	62.6	68.9	15.9	13.8	0.95	0.94
PFCN	6.3	6.2	26.0	25.8	30.2	31.1	0.96	0.96
CS-NN	6.0	6.5	25.3	25.5	31.8	30.8	0.93	0.92
BDCF	**5.7**	**5.9**	**23.1**	**24.9**	**34.3**	**32.5**	**0.98**	**0.97**



(a) Simulated LR HS (b) Simulated HR MS (c) Reference image (d) CNMF (e) FUSE
(f) HCM (g) SFIM (h) GSA (i) Two-CNN-Fu (j) BDCF

Fig. 8. Fusion results in Poyang Lake-1 data set from different methods. (a) Simulated LR HS. (b) Simulated HR MS. (c) Reference image. (d) CNMF. (e) FUSE. (f) HCM. (g) SFIM. (h) GSA. (i) Two-CNN-Fu. (j) BDCF.

fluctuations in all methods although the trend of these curves is correct, BDCF achieves the most approximate results and the differences mainly exists between 700 and 1249 nm, and the fluctuations of Two-CNN-Fu are the biggest. In pond area [see Fig. 13(d)], there are wrong peaks in FUSE around 420–642 nm. Besides, the fluctuations in curves of HCM, CNMF

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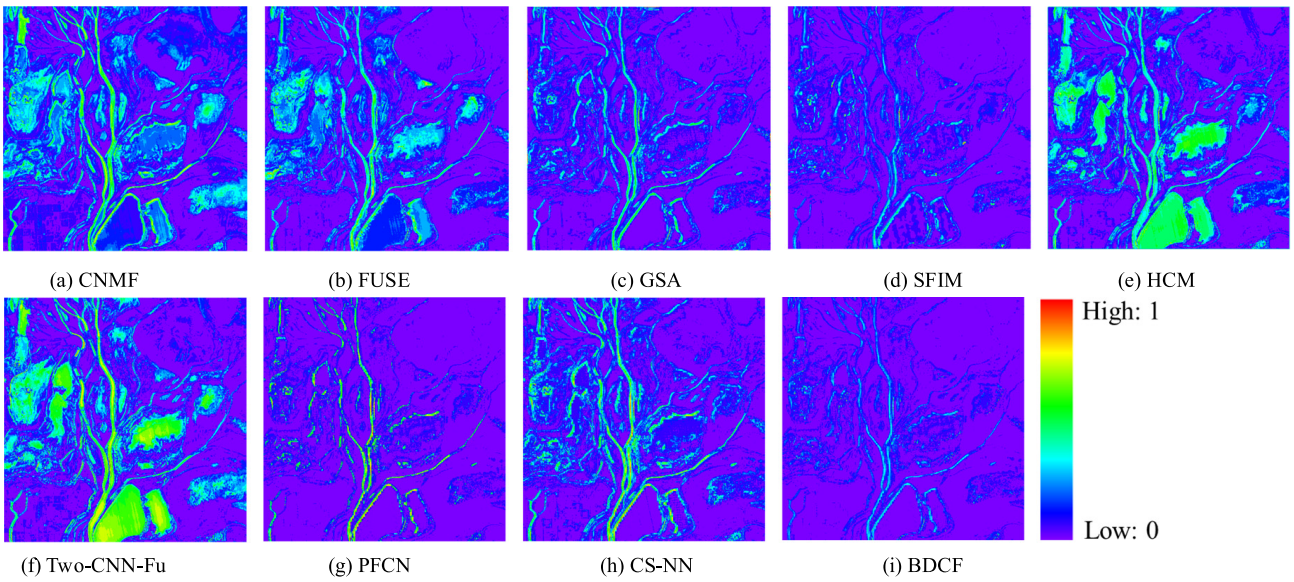


Fig. 9. SAM map in Poyang Lake-1 data set. (a) CNMF. (b) FUSE. (c) GSA. (d) SFIM. (e) HCM. (f) Two-CNN-Fu. (g) PFCN. (h) CS-NN. (i) BDCF.

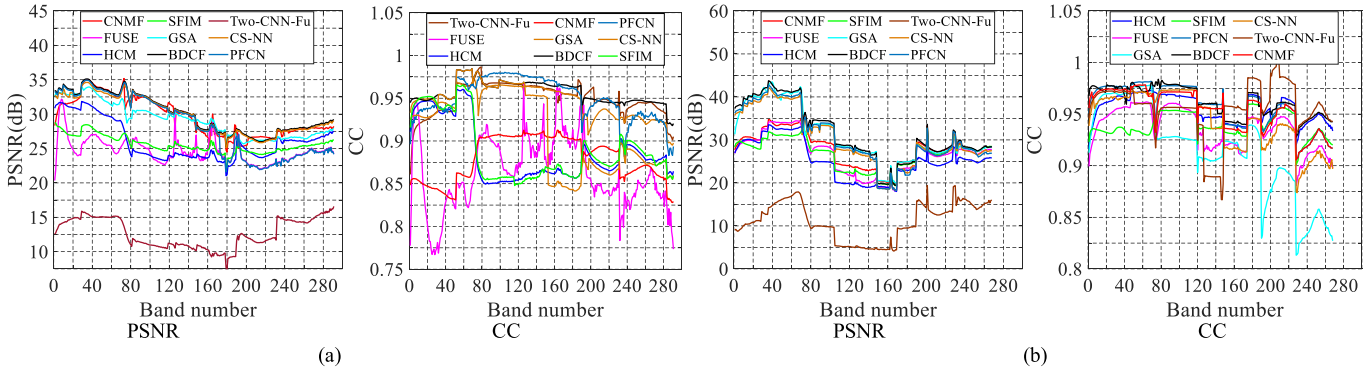


Fig. 10. PSNR and CC of all bands. (a) Langfang. (b) Poyang Lake-1.

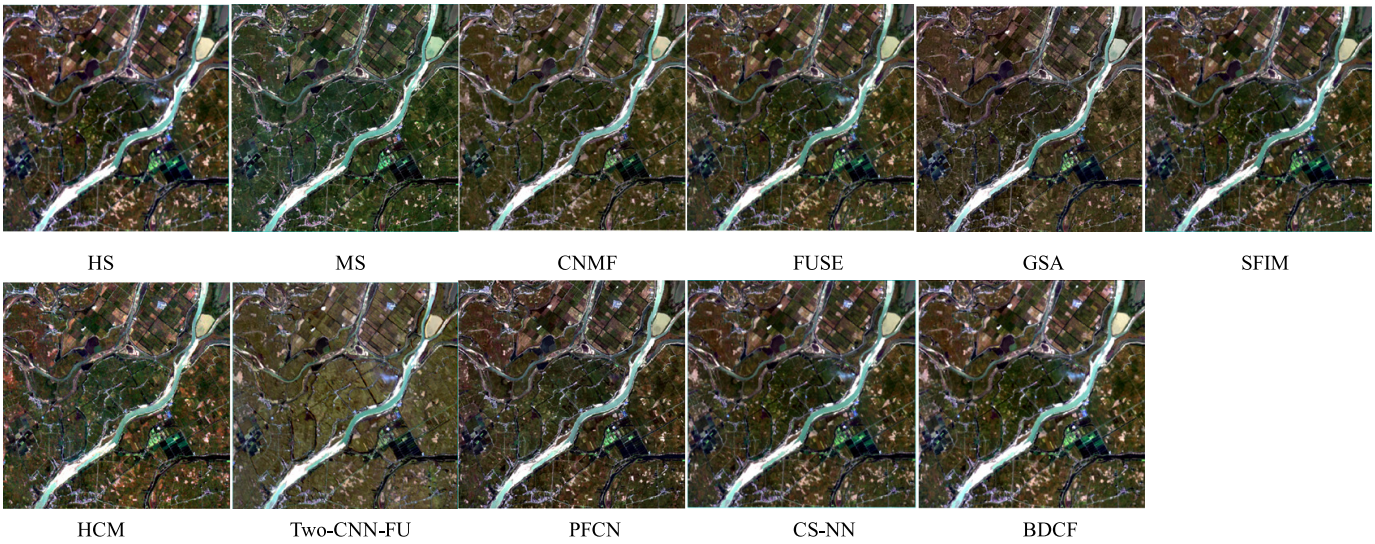


Fig. 11. Fusion results in real Poyang Lake-2 data set.

and GSA are clear, and BDCF obtained the best spectral curve. In Taihu Lake data set, for the river, vegetation and pond area [see Fig. 14(a)–(c)], the curve of BDCF is almost consistent with the reference spectral curve, however, Two-CNN-Fu has a wrong peak value in the river of 640 nm, HCM has a wrong trend of spectral curve in the river of 1502–1754 nm, the curves of Two-CNN-Fu, CNMF, FUSE CS-NN and HCM fluctuate severely in the river of 2000–2300 nm, the curves of benchmark methods are completely wrong in vegetation of 390–514 nm and have a wrong peak value in the pond

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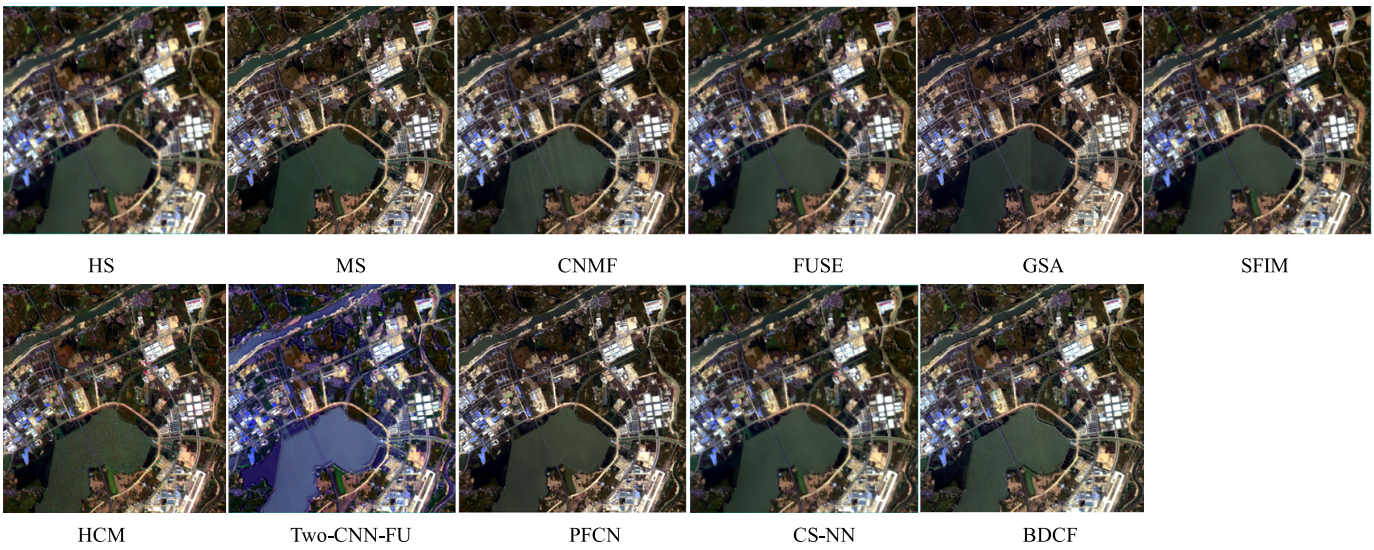


Fig. 12. Fusion results in Taihu Lake data set.

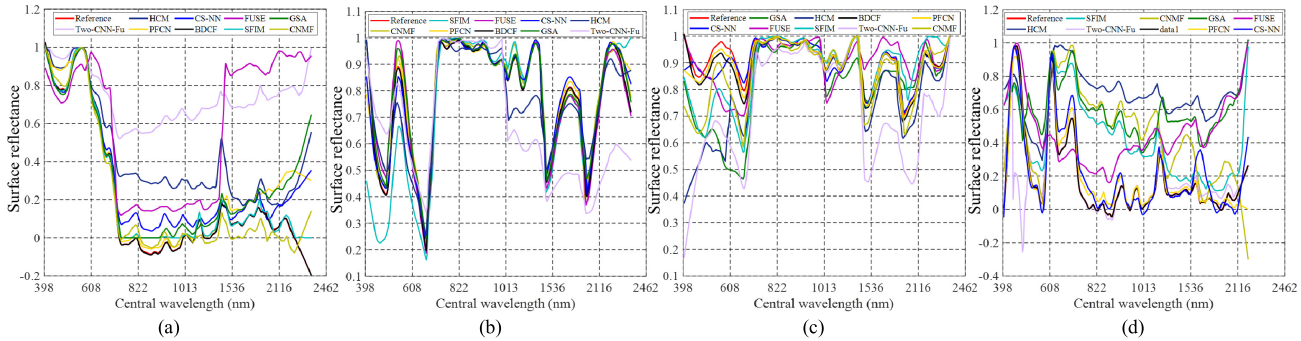


Fig. 13. Spectral differences of different objects in Poyang Lake-2 data set. (a) River. (b) Vegetation. (c) Bare land. (d) Pond.

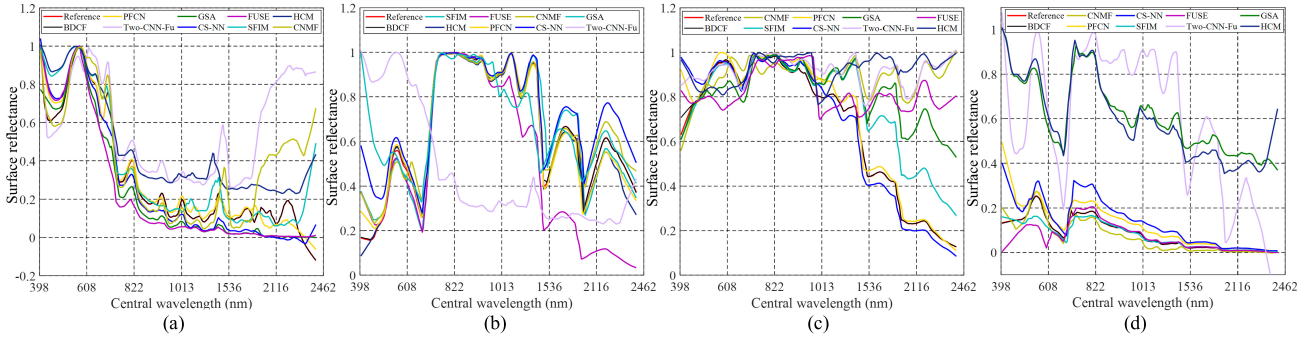


Fig. 14. Spectral differences of different objects in Taihu Lake data set. (a) River. (b) Vegetation. (c) Bare land. (d) Pond.

of 1502 nm. In bare land [see Fig. 14(c)], there are obvious fluctuations in the spectral curve of all methods compared with the reference spectral curve; however, our method is still superior to other methods and obtains the most approximate value.

D. Discussion

MS/HS fusion performance is limited by fundamental nonoverlapped spectrum range between HS and MS images. In addition, these fusion methods have some shortcomings:

1) The CS-based MS/HS fusion methods simply construct the intensity component of HS data and the weight matrix that inject the detail information into HS data. Accordingly, they generally have high computational efficiency for engineering applications; however, some spectral characteristics of HS data are changed during the CS, and they are generally limited by spectral fidelity [45].
2) The MRA-based MS/HS fusion methods are based on injection of the high spatial detail information of MS data, to preserve the spectral information of HS, they generally achieve better spectral fidelity and high computational efficiency; however, some methods are limited in spatial sharpening effect.
3) The unmixing-based methods and Bayesian-based methods generally obtain fused image based on iterative

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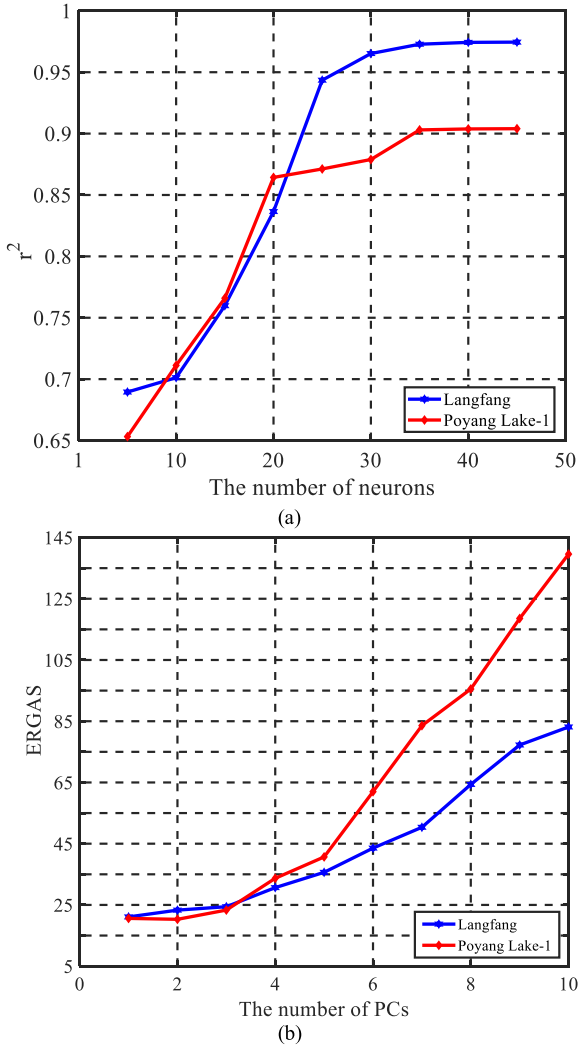


Fig. 15. Influence of different parameters on network. (a) Number of neurons. (b) Number of PCs when the number of neurons is 30.

optimization solution, besides, several model parameters are existed, to make these methods with low calculation efficiency.

4) The deep learning-based methods are based on the supervised network training, based on large numbers of data sets. However, on the one hand, the existing opening HS and MS data are limited; on the other hand, the performance is limited by the pretrained network, and for different tasks, the network are generally retrained.

In this article, we proposed a novel BDCF for MS/HS fusion with both high calculation efficiency and high fidelity. The proposed BDCF is simple yet effective, with high computational efficiency and few parameters, and it achieved comparative fusion results compared with other state-of-the-art methods.

The proposed method includes two simple parameters: the number of neurons and the number of PCs. The setting of parameters was investigated by two simulated data sets (see Fig. 15). There is a tradeoff between the number of neurons and computational efficiency. The increase of neurons improves the performance of training; however, the computational burden of the proposed method is increasing; besides,

TABLE III
EVALUATION RESULTS OF DIFFERENT METHODS ON OPERATION TIME

Methods	Times (S)			
	Langfang	Poyang Lake-1	Poyang Lake-2	Taihu Lake
CNMF	51.36	68.25	321.14	364.21
FUSE	42.59	50.36	301.56	316.17
Two-CNN-Fu	149.41	221.32	1504.31	1824.32
PFCN	152.39	236.15	1589.38	1864.11
CS-NN	50.49	69.46	295.24	312.01
BDCF	**32.34**	**46.59**	**90.57**	**102.34**

the rise of r^2 tends to be flat when the number of neurons is greater than 30. Therefore, in these experiments, the optimal number of neurons should be set to 30; however, it can also be appropriately increased when this image is large to ensure better performance of training. The number of PCs is inversely proportional to the quality of data fusion. The less PCs are used, the better the fusion quality is. The reason is that the more PCs are used, the more components need to be replaced, and the greater the spectral distortion, Therefore, the optimal number of PCs is set to 1 or 2 in these experiments.

BDCF was implemented with MATLAB 2018a on an Intel Core i5 processor and 12-GB RAM. Computation times comparison between our method and the unmixing-based methods, Bayesian-based methods and deep learning-based methods on both four data sets are shown in Table III. CNMF, FUSE, Two-CNN-Fu, PFCN, and CS-NN have relatively low calculation efficiency. Our method achieves good computational performance.

IV. CONCLUSION

In this article, a novel BDCF HS and MS fusion method was proposed. The HS data were divided to overlapped bands and nonoverlapped bands according to the spectral rang of data set. For the overlapped bands, an improved CS-based method was proposed by combining neural network framework. In the nonoverlapped bands, the relationship between overlapped and nonoverlapped bands was trained by the neural network, and the well-trained relationship was then mapped to the fused overlapped bands to predict the fused nonoverlapped bands. The proposed method not only obtained comparative result in overlapped bands, but also improved the spectral fidelity of the nonoverlapped bands. We verified the proposed method on two simulated data sets and two real data sets. Based on visual and quantitative results, our method achieved better spectral fidelity and obtained clearer spatial details compared with the other state-of-the-art methods.

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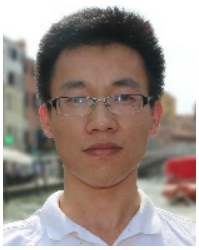
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